

Instructions:

1. Navigate to documents/
2. Place image to trace in image folder.
3. Ensure usb is connected to computer.
4. Type `sudo chmod 777 /dev/ttyACM0`, this give the program the ability to write to the motor.
5. Type `python3 ./run_motor`
6. Enter scale, standard fit has image fill a 255 x 255 mm square centered in the center of the circle. The 0-1 scale is just a multiplicative factor.
7. Enter amount of points you would like to assign to the motor, the recommended points gives the amount of points before new “filler” points must be created.
8. The plot points prompt is to check that the correct shape has been traced to the users liking. If you plot points do not close out of the new window after, this will crash the program.
9. Select which mode you would like to use, Pausing moves from one point to another while linear smoothly traces the shape. Circular is Michael's old code, functions similar to pausing but specifically for circles.
10. Type go when you are ready to begin the run.
11. In linear mode, after the run is complete you will be prompted as to whether you want to run again.

Notes:

1. Place the motor in the corner nearest to the power source before starting.
2. Run “py resetmot” to return the motor to 0, 0 (0, 0 is the location of the motor on initial startup).
3. Run “py centermot” to center the motor in the middle of the sensor.